INCH-POUND

A-A-55157

11 June 1992

COMMERCIAL ITEM DESCRIPTION

# RESISTOR, FIXED, WIREWOUND, INDUCTIVE, POWER TYPE

The General Services Administration
has authorized the use of
this commercial item description.

1. Abstract. This commercial item description (CID) covers the general requirements for wirewound, power type, resistors. These resistors are an all welded construction and rugged, flame resistant conformal vitreous enamel coating to ensure successful performance under high temperatures. Resistors supplied to this (CID) have hollow core construction which accommodates rigid mounting with brackets or through bolts.

1.1 Application. Resistors covered by this CID are intended for commercial/industrial applications and shall not be used in military systems needing stringent environmental and electrical requirements.

2. Part or identifying number (PIN). The PIN for the CID shall be as shown in the following example:

| A-A-55157 | - | XXX | K |
|---|---|---|---|
| CID number | | Resistance value (see 3.1) | Resistance tolerance (See 3.2) |

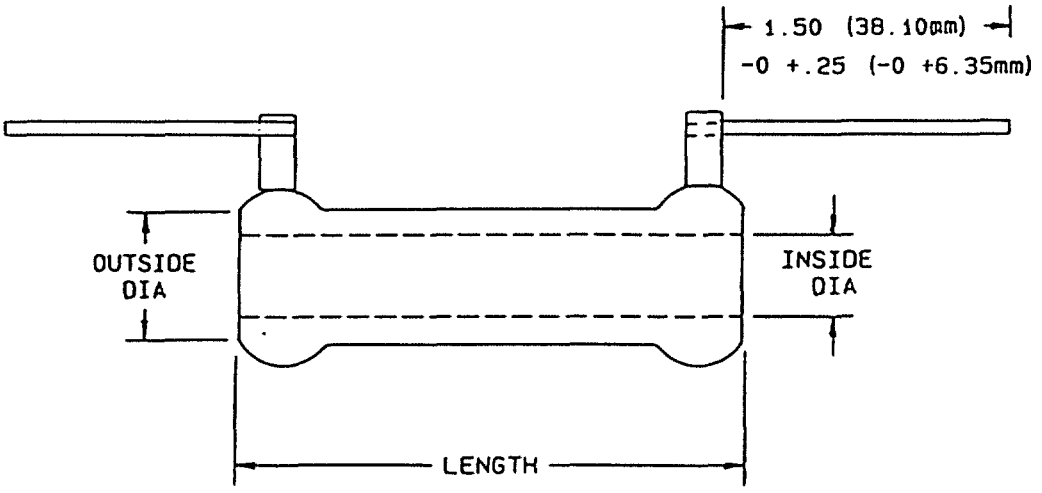
3. Salient characteristics.

3.1 Resistance. The nominal resistance expressed in ohms is identified by three characters, consisting of two digits and a letter. The letter is used simultaneously as a decimal point and a multiplier. For resistance values greater than or equal to 1 ohm but less than 1 kilohm, the letter "R" is used to represent a decimal point, greater than or equal to 1 kilohm but less than 1 megohm, the letter "K" is used to represent a decimal point. All digits preceding and following the letters "R" and "K" of the group represent significant figures. Values between and including 100 and 990 ohms are represented by the value itself. The following are examples of the resistance values:

1R0 = 1 ohm
10R = 10 ohms
100 = 100 ohms
1K0 = 1,000 ohms
12K = 12,000 ohms


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| Core Dimensions | | | | | | |
|---|---|---|---|---|---|---|
| Length | | outside diameter | | inside diameter | | Leads gauge |
| inch | mm | inch | mm | inch | mm | |
| 1.000 | 25.40 | 0.313 | 7.94 | 0.188 | 4.76 | 18 |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 1. Configuration and dimensions.



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3.2 Resistance value. The resistance values shall be 1 ohm to 25,000 ohms.

3.3 Resistance tolerance. Resistors are available in resistance tolerance (J) ±5 percent.

3.4 Temperature coefficient. The temperature coefficient shall be ±400 ppm < 5Ω, ±260 ppm ≥ 5Ω.

3.5 Maximum voltage. The maximum voltage rating shall be 250 volts.

3.6 Power rating. The power rating shall be 8.0 watts at 25°C derated to zero power at 350°C (see figure 2). When the resistance exceeds 2,500 ohms, derate power rating by 25 percent to improve reliability.

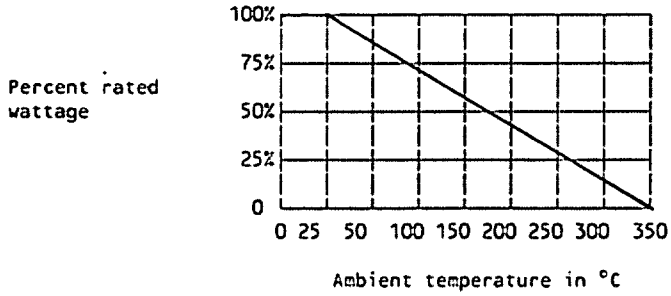



FIGURE 2. Derating curve.

3.7 Overload. Resistors shall be able to withstand 10 times rated wattage for 5 seconds.

3.8 Weight. The maximum weight shall be 5.5 grams.

3.9 Design and construction. Resistors supplied to this commercial item description shall have an all welded construction and rugged, flame resistant conformal vitreous enamel coating to ensure successful performance under high temperatures.

3.10 Marking. Resistors supplied to this commercial item description shall be marked with the manufacturers standard commercial PIN.

4. Quality assurance provisions.

4.1 Responsibility for inspection. The contractor is responsible for the performance of all inspections, examinations, and test requirements specified in the contract or purchase order. The contractor may use his own or any other facilities suitable for inspection requirements specified in the contract or purchase order, unless otherwise disapproved by the procuring activity. The acquiring activity reserves the right to perform any of the inspections set forth in the contract or purchase order where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Contractor certification statement. The contractor shall certify and maintain objective quality evidence that the product offered meets this CID, and that the product conforms to the producers own drawings, specifications, standards, quality assurances practices, and is the same as the product provided as a bid sample. The acquiring activity reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.3 Certificate of compliance. A certificate of compliance shall accompany all resistors supplied to this CID.

5. Packaging.

5.1 Packaging. Preservation, packaging, labeling, and marking shall be as specified in the contract or purchase order.



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6. Notes. This section contains relevant information which is useful to buyers, users, and suppliers in the process of acquiring the item.

6.1 Ordering data. The contract or purchase order shall specify the following:

a. CID document number and revision and CID PIN.
b. Quality assurance provisions.
c. Packaging requirements.

6.2 Suggested sources of supply. Suggested source supply are listed herein. Additional sources will be added as they become available.

Suggested source of supply.

| Vendor commercial PIN 1/ | Vendor CAGE number |
|---|---|
| B8J*** | 44655 |

1/ Asterisks represent coded resistance values see 3.1 and 3.2.

Vendor CAGE number
44655

Vendor name and address
Ohmite Manufacturing
3601 Howard Street
Skokie, IL 60076

CONCLUDING MATERIAL

Civil Agency Coordinating Activity:
GSA - 7FXE

Preparing activity:
DLA-ES

(Project 5905-D323)



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